

National Institute of Technology Hamirpur
Department of Computer Science and Engineering
CS-414 Information Security and Privacy Lab
Assignment- 2

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Submission Deadline: 14/08/2026

Session: Odd Sem, 2026-27

Date: 07 Aug. 2026

- 1. Write a program to find the linear Diophantine equation (LDE) solutions (Particular solution and General Solution) for the given equations:**

a. $25x + 10y = 15$

b. $19x + 13y = 20$

c. $14x + 21y = 77$

d. $40x + 16y = 88$

Linear Diophantine equation:

- A linear Diophantine equation of two variables is $ax + by = c$.
- **One immediate application of extended euclidean algorithm** is to find the solutions to the linear Diophantine equations of two variables, an equation of type $ax + by = c$. We need to find integer values for x and y that satisfy the equation.
- This type of equation has **either no solution or an infinite number of solutions**.
- Let $d = \gcd(a, b)$. If $d \nmid c$, then the equation has no solution. If $d \mid c$, then we have an infinite number of solutions. One of them is called **the particular** solution the rest, general solution.

▪ Particular Solution

If $d \mid c$, a particular solution to the above equation can be found using the following steps:

1. Reduce the equation to $a_1x + b_1y = c_1$ by dividing both sides of the equation by d . This is possible because d divides a , b , and c by the assumption.
2. Solve for s and t in the relation $a_1s + b_1t = 1$ using the extended Euclidean Algo.
3. The particular solution can be found:

Particular solution: $x_0 = (c/d)s$ and $y_0 = (c/d)t$

▪ General Solutions

After finding the particular solution, the general solutions can be found:

$x = x_0 + k(b/d)$ and $y = y_0 - k(a/d)$

- Find the particular and general solutions to the equation $21x + 14y = 35$

• Step 1: Find 'd',
 $d = \gcd(21, 14) = 7$

Now, divide the equation by 7

$$21x + 14y = 35$$

$$\Rightarrow 3x + 2y = 5$$

- Step 2: Find the value of 's' and 't' using Extended Euclidean algorithm.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
1	3	2	1	1	0	1	0	1	-1
2	2	1	0	0	1	-2	1	-1	3
	1	0		1	-2		-1	3	

$$s = 1 \quad \text{and} \quad t = -1$$

- Calculate the "particular Solution":

$$\begin{aligned} x_0 &= \left(\frac{c}{d}\right)s & \text{and} & & y_0 &= \left(\frac{c}{d}\right)t \\ &= \left(\frac{35}{7}\right)(1) & & & &= \left(\frac{35}{7}\right)(-1) \\ &= 5 & & & &= -5 \end{aligned}$$

- Calculate the "General Solution":

$$\begin{aligned} x &= x_0 + k\left(\frac{b}{d}\right) & \text{and} & & y &= y_0 - k\left(\frac{a}{d}\right) \\ &= 5 + k(2) & & & &= -5 - k(3) \\ \underline{x = 5 + 2k} & & & & \underline{y = -5 - 3k} \end{aligned}$$

where, k is an integer,

so, $(5, -5), (7, -8), (9, -11), \dots$ so on